E1.01 SOLUTIONS

Solution 1

a) KCL at X gives: $\frac{X-13}{4}+\frac{X-(Y-2)}{3}+\frac{X-Y}{1}=0$, or $19X-16Y-31=0\ldots.(1)$

KCL for super-node including everything except LH source, 4Ω & 2Ω:

$$\frac{13-X}{4}=\frac{Y-2}{1}, \text{ or } X+4Y-21=0\ \ldots.(2)$$

Solving (1) and (2) gives X=5, Y=4. [5]

b) With the current source set to zero, we have a potential divider with 3 Ω at the top and 6 Ω at the bottom and a source voltage of 12 V. We therefore have $X_V=\frac{6}{9}\times 12=8.$

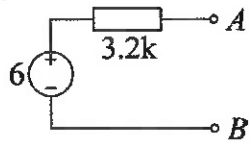
With the voltage source set to zero, we have a current source driving 5//(1 + 3) Ω. The current in the branch containing the 3 Ω and the 1 Ω is $\frac{5}{9}\times 3=\frac{5}{3}$ A, so the voltage at X is $X_I=\frac{5}{3}\times 3=5$.

By superposition the voltage at X in the presence of both sources is $X=X_V+X_I=13$. [5]

c) The current source sees a load of 6k//(10k + 4k) when the terminals are open-circuit. The voltage developed across this load is $\frac{6\times 14}{6+14}\times 5=21$. The Thévenin voltage is the fraction of this voltage that appears across the 4k, so $V_{TH}=21\times\frac{4}{14}=6$.

To find the Thévenin resistance we set the source to zero and measure the resistance between A and B. This gives $R_{TH}=4k//(10k+6k)=\frac{4\times 16}{4+16}=\frac{16}{5}=3.2k.$

The Thévenin equivalent is therefore:



[5]

d) The simplest solution is by superposition. The input U_1 sees an inverting amplifier with a gain of –R/(R/5) = –5. The input U_2 passes through an attenuator with a gain of 1/3 and then sees a non-inverting amplifier with a gain of +(R + R/5)/(R/5) = +6. The overall gain seen by U_2 is therefore +6/3 = +2. Combining these results the expression for Z is:

$$Z=-5U_1+2U_2$$ [5]

e) The impedance of the inductor at 50 Hz is $Z_L = j\omega L = 31.42j$. The combined impedance of the parallel RL combination is $Z_{T=} \dfrac{27 \times 31.42j}{27 + 31.42j} = 15.53 + 13.35j$. The RMS voltage across the parallel RL is then $\tilde{V}_T = \dfrac{Z_T}{1 + Z_T}\tilde{V}_S$. With $\tilde{V}_S = 230$, $\tilde{V}_T = 221.7\angle 1.75° = 221.6 + j6.77$.

The source current is $\tilde{I}_S = \dfrac{-\tilde{V}_S}{1 + Z_T} = 10.84\angle 141.07 = -8.42 + 6.80j$, where we have used passive sign convention.

For the resistors $S = P + j0 = \dfrac{\tilde{V}^2}{R} = \tilde{I}^2 R$, while for the inductor $S = 0 + jQ = \dfrac{j\tilde{V}^2}{\omega L}$.

For the source we can use $S = \tilde{V}_S \times \tilde{I}_S^*$. Using the above equations the complex powers are:

1 Ω resistor: S = P = 117 W

27 Ω resistor: S = P = 1820 W

Inductor: S = jQ, where Q = 1564 VAR

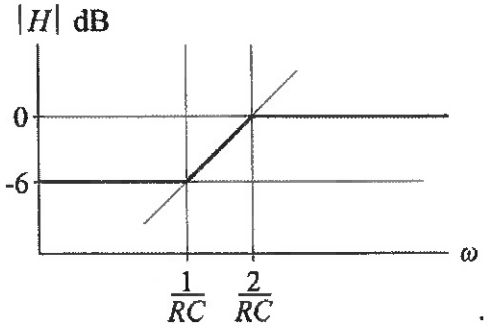
Source: S = P + jQ, with P = -1937 W, Q = -1564 VAR

Sanity check: P values and Q values separately sum to zero. [5]

f) The circuit is a potential divider where the lower element is a resistor and the upper element is a parallel RC which has an impedance of $Z_{RC} = R/(1 + j\omega RC)$. The transfer function is therefore:

$$H = \frac{Y}{X} = \frac{R}{R + R/(1 + j\omega C)} = \frac{1 + j\omega RC}{2 + j\omega RC}$$

The low-freq asymptote is $H = 0.5$, the high-freq asymptote is $H = 1$, and there are two corner frequencies at $\omega = 1/(RC)$ and $\omega = 2/(RC)$.



[5]

g) If we call W the output of the F block, then we have:

$$W = F(X - Y) \quad ; \quad Y = G(W - HY)$$

Combining these to eliminate W gives:

$$Y = G[F(X - Y) - HY]$$

$$\Rightarrow Y(1 + GF + GH) = GFX$$

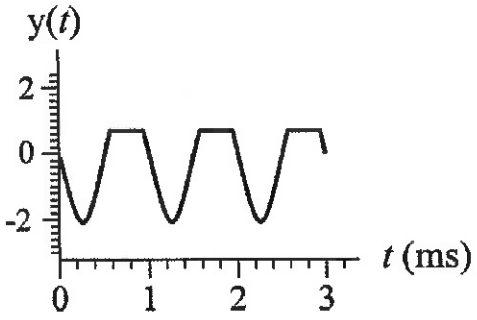
$$\Rightarrow \frac{Y}{X} = \frac{GF}{1 + GF + GH}$$

h) The circuit has negative feedback so we can assume the inverting input of the opamp is a virtual earth.

If $y(t) < 0.7$ the diode will be off and the circuit will behave as an inverting amplifier with a gain of -1. This will apply when $x(t) > -0.7$.

When $x(t) < -0.7$, the diode will turn on and clamp the output at $y(t) = 0.7$.

The 1 kHz sine wave input with amplitude 2 V will therefore generate the output shown below.



Solution 2

a) The amplifier has unity gain, so the phasor at the + input is just Y. If we label the junction between R, mR and nC as A, then we have the following KCL equations:

$$\frac{A-Y}{R} = j\omega CY\,, \;\Rightarrow\; A = Y(1+j\omega RC) \quad \ldots . (1)$$

$$\frac{X-A}{mR} + \frac{Y-A}{R} + j\omega nC(Y-A) = 0\,, \;\Rightarrow\; (X-A) + m(Y-A)(1+j\omega nCR) = 0 \quad \ldots (2)$$

Using (1) to eliminate A and $(Y-A)$ from (2) gives:

$$X - Y(1+j\omega RC) - mj\omega RCY(1+nj\omega RC) = 0$$

which, after rearrangement, gives the required result. [8]

b) The standard form for the transfer function is given in the *Information for Candidates*. Comparing coefficients we see that:

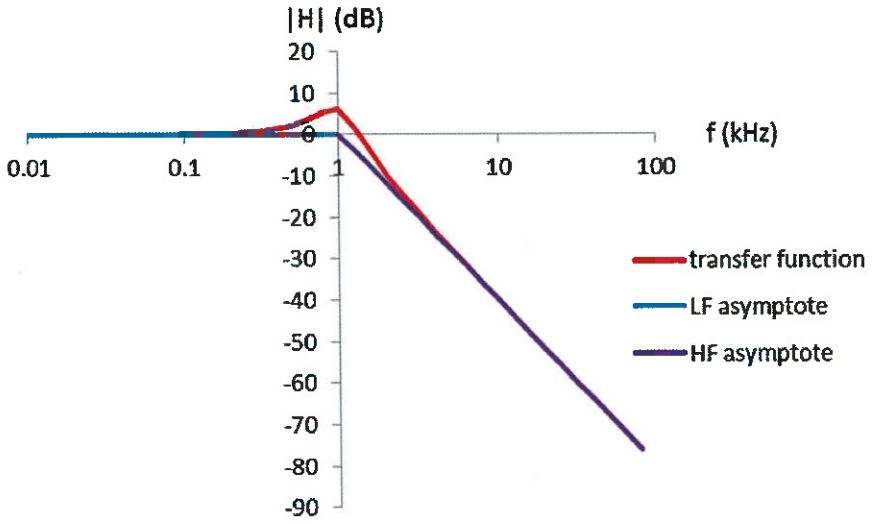
$$mn(RC)^2 = \omega_0^{-2} \;\Rightarrow\; \omega_0 = \frac{1}{\sqrt{mn}RC}$$

$$(m+1)RC = \frac{2\zeta}{\omega_0} \;\Rightarrow\; Q = \frac{1}{2\zeta} = \frac{1}{(m+1)RC\omega_0} = \frac{\sqrt{mn}}{(m+1)} \qquad [6]$$

c) We require $\omega_0 = 2000\pi$ rad/s, and we have $m = 1$, $n = 16$, $C = 10$ nF. Rearranging the given formula for ω_0, we find $R = 3.98$ kΩ. [4]

The low-freq and high-freq asymptotes are $|H| = 1$ and $|H| = (\omega_0/\omega)^2$. On the magnitude response plot these are straight lines with slopes of 0 and -40 dB/decade, crossing at the point (1 kHz, 0 dB).

The filter has Q = 2, so there is a small resonance peak, reaching +6 dB at the corner frequency.



[6]

d) In the case $m = n = 1$, the transfer function becomes:

$$H(j\omega) = \frac{1}{(j\omega RC)^2 + 2j\omega RC + 1} = \frac{1}{(j\omega RC + 1)(j\omega RC + 1)}$$

From the right-hand expression it is clear that the filter is equivalent to two cascaded RC filters as shown in Figure 2.2. The buffer is needed so that the second filter does not load the first one and modify its transfer function.

One reason that the Sallen-Key might be preferred is that it always maintains a low output impedance, the output coming directly from the opamp. [6]

Solution 3

a) The circuit is a potential divider where the upper element is a resistor and the lower element is a series RC. The transfer function is therefore:

$$H = \frac{Y}{X} = \frac{R + 1/j\omega C}{(k-1)R + R + 1/j\omega C} = \frac{1 + j\omega RC}{1 + j\omega kRC} = \frac{1 + ju}{1 + jku}$$

It follows that:

$$|H| = \frac{|1 + ju|}{|1 + jku|} = \sqrt{\frac{1 + u^2}{1 + k^2u^2}}$$

$$\angle H = \angle(1 + ju) - \angle(1 + jku) = \arctan(u) - \arctan(ku) \qquad [4]$$

b) $\angle H$ has a single minimum for $k > 1$ which we can find by differentiating:

$$\frac{\partial \angle H}{\partial u} = \frac{1}{1+u^2} - \frac{k}{1+k^2u^2} = \frac{(1-k)(1-ku^2)}{(1+u^2)(1+k^2u^2)}$$

We can see that $\angle H = 0$ when $1 - ku^2 = 0$ or $u = \frac{1}{\sqrt{k}}$. Substituting this value of u into the expressions for $|H|$ and $\angle H$:

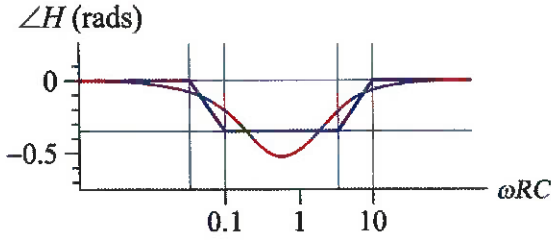
$$|H| = \sqrt{\frac{1 + 1/k}{1 + k}} = \frac{1}{\sqrt{k}}$$

$$\angle H = \arctan(\tfrac{1}{\sqrt{k}}) - \arctan(\sqrt{k}) = \arctan\left(\frac{\frac{1}{\sqrt{k}} - \sqrt{k}}{2}\right) = -\arctan\left(\frac{k-1}{2\sqrt{k}}\right)$$

where we have used the formula for tan(A – B) to obtain the second expression. [8]

c) For a filter with $k = 3$, the terms in the numerator and denominator of the transfer function will have corner frequencies of $1/(RC)$ and $1/(3RC)$ respectively. The phase response plot will therefore have corners at $[0.1/(RC)]^+$, $[10/(RC)]^-$, $[0.1/(3RC)]^-$, $[10/(3RC)]^+$, where the sign indicates an increase (+) or decrease (–) in the gradient of the plot. Placing the corners in order, we have: $[0.1/(3RC)]^-$, $[0.1/(RC)]^+$, $[10/(3RC)]^+$, $[10/(RC)]^-$.

The phase at low frequency is zero. The gradient between the first two corners is $-(\pi/4)$/decade, and their separation is log(3) = 0.477 decades, so the phase in the middle section is $-0.477 \times \pi/4 = -0.375$ rads = $-21.5°$. This is the minimum phase shift according to the straight line approximation. [The actual minimum phase shift, from the formula in part b), is $-\arctan\left(1/\sqrt{3}\right) = -\pi/6$.]



[6]

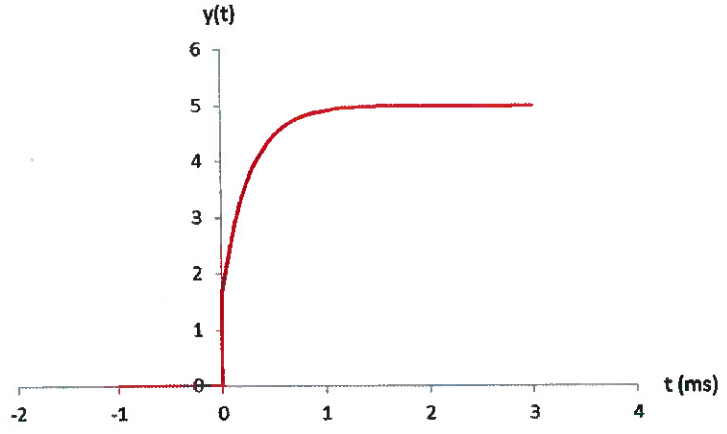
d) For a DC input, the steady state condition is $y = x$, so the steady state outputs are $y_{ss} = 0$ for $t < 0$ and $y_{ss} = 5$ for $t > 0$. For $t < 0$ we assume steady state conditions so that $y = y_{ss}$.

The capacitor voltage is zero for $t = 0^-$, and it cannot change instantaneously, so it is still zero at $t = 0^+$. The output voltage at $t = 0^+$ is therefore $y(0^+) = 5/3$ (potential divider).

Since $y_{ss} = 5$, and $y(0^+) = 5/3$, the initial amplitude of the transient component must be $(5/3 - 5) = -10/3$, and the full expression for $y(t)$ is:

$$y(t) = \begin{cases} 0 & t < 0 \\ 5 - 3.33e^{-t/\tau} & t \geq 0 \end{cases}$$

The time constant is $\tau = kRC = 275$ µs.



[6]

e) The steady state output for $t < 0$ is $y_{ss} = 0$ as before. For $t > 0$ we need the steady-state response of the circuit to the sinusoidal input x(t) = 5sin(2000πt). The signal frequency corresponds to the minimum phase point for the filter, so we know from part b) that $|H| = 1/\sqrt{3}$ and $\angle H = -\pi/6$. Therefore $y_{ss}(t) = \frac{5}{\sqrt{3}}\sin(2000\pi t - \pi/6)$. There is no step change in the input this time, so $y(0^+) = 0$ and the initial amplitude of the transient component must be $-y_{ss}(0) = -\frac{5}{\sqrt{3}}\sin(-\pi/6) = \frac{5}{2\sqrt{3}}$. The full expression for $y(t)$ is then:

$$y(t) = \begin{cases} 0 & t < 0 \\ \frac{5}{\sqrt{3}}\sin(2000\pi t - \pi/6) + \frac{5}{2\sqrt{3}}e^{-t/\tau} & t \geq 0 \end{cases}$$

[6]